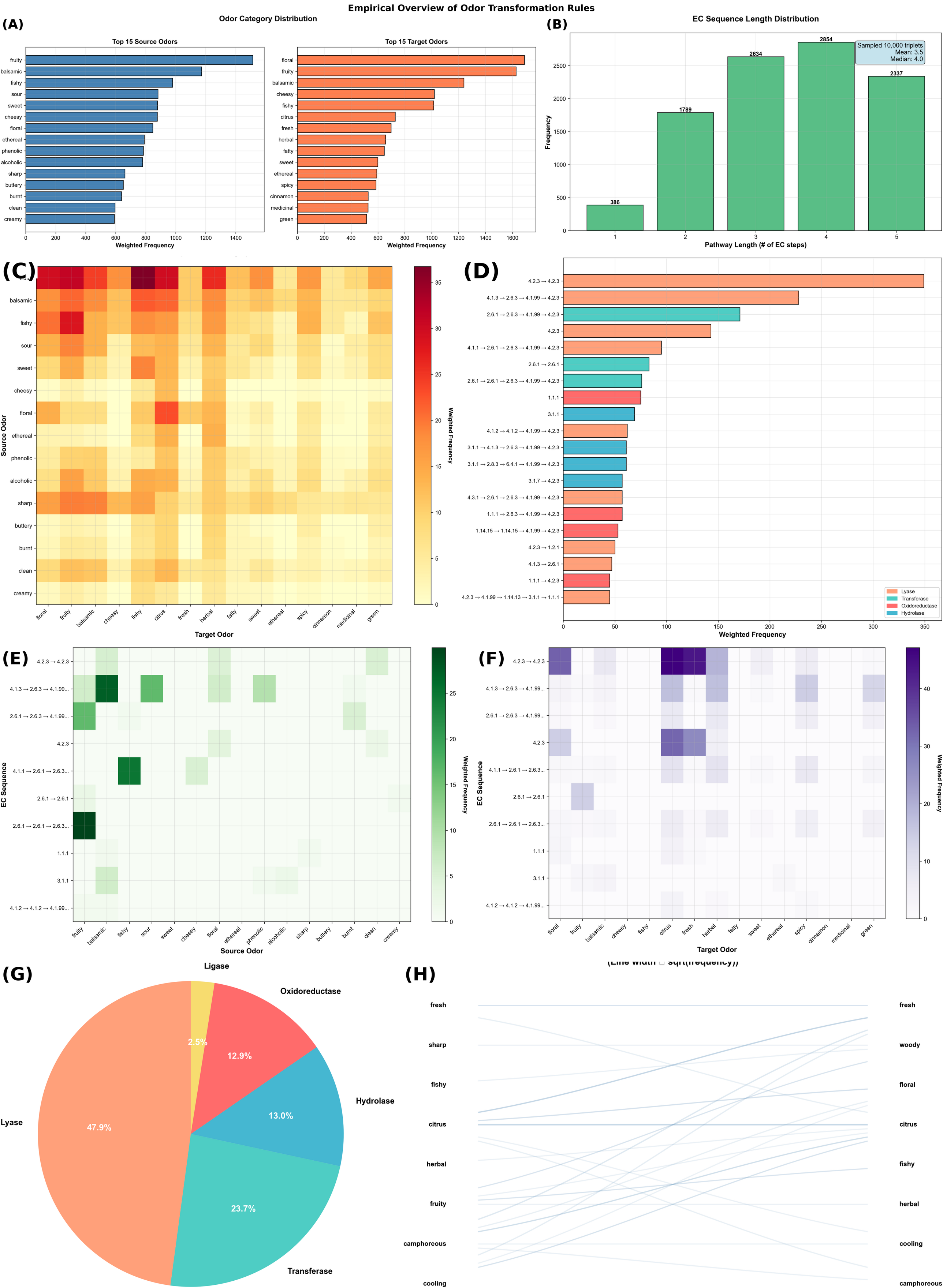
Empirical Overview of Odor Transformation Rules
(A)
Odor Category Distribution
Top 15 Source Odors
fruity
balsamic
fishy
sour
sweet
cheesy
floral
ethereal
phenolic
alcoholic
sharp
buttery
burnt
clean
creamy
Weighted Frequency
Top 15 Target Odors
floral
fruity
balsamic
cheesy
fishy
citrus
fresh
herbal
fatty
sweet
ethereal
spicy
cinnamon
medicinal
green
Weighted Frequency
(B)
EC Sequence Length Distribution
Sampled 10,000 triplets
Mean: 3.5
Median: 4.0
386
1789
2634
2854
2337
Frequency
Pathway Length (# of EC steps)
(C)
Source Odor
Target Odor
Weighted Frequency
(D)
4.2.3 → 4.2.3
4.1.3 → 2.6.3 → 4.1.99 → 4.2.3
2.6.1 → 2.6.3 → 4.1.99 → 4.2.3
4.2.3
4.1.1 → 2.6.1 → 2.6.3 → 4.1.99 → 4.2.3
2.6.1 → 2.6.1
2.6.1 → 2.6.1 → 2.6.3 → 4.1.99 → 4.2.3
1.1.1
3.1.1
4.1.2 → 4.1.2 → 4.1.99 → 4.2.3
3.1.1 → 4.1.3 → 2.6.3 → 4.1.99 → 4.2.3
3.1.1 → 2.8.3 → 6.4.1 → 4.1.99 → 4.2.3
3.1.7 → 4.2.3
4.3.1 → 2.6.1 → 2.6.3 → 4.1.99 → 4.2.3
1.1.1 → 2.6.3 → 4.1.99 → 4.2.3
1.14.15 → 1.14.15 → 4.1.99 → 4.2.3
4.2.3 → 1.2.1
4.1.3 → 2.6.1
1.1.1 → 4.2.3
4.2.3 → 4.1.99 → 1.14.13 → 3.1.1 → 1.1.1
Lyase
Transferase
Oxidoreductase
Hydrolase
Weighted Frequency
(E)
EC Sequence
Source Odor
Weighted Frequency
(F)
EC Sequence
Target Odor
Weighted Frequency
(G)
Ligase
Oxidoreductase
Hydrolase
Transferase
Lyase
2.5%
12.9%
13.0%
23.7%
47.9%
(H)
(Line width ∝ sqrt(frequency))
fresh
sharp
fishy
citrus
herbal
fruity
camphoreous
cooling
fresh
woody
floral
citrus
fishy
herbal
cooling
camphoreous